

Operating Systems

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Report:

Compile:

```
gcc -Wall -o prog prog.c
```

Run:

```
./prog input.1 FCFS
```

```
./prog input.1 RR
```

```
./prog input.1 SJF
```

Screenshots:

SJF

```

matthew@LAPTOP-9I9622SL:~/CS4420/instruction-cs4420/Homework 4$ ./prog input.1 SJF
SJF:
Time 0: PID 0 running
Time 1: PID 0 running
Time 2: PID 0 running
Time 3: PID 0 running
Time 4: PID 0 running
Time 5: PID 0 running
Time 6: PID 0 running
Time 7: PID 0 running
Time 8: PID 0 running
Time 9: PID 0 running
Time 10: PID 0 running
Time 11: PID 0 running
Time 12: PID 0 finished
Time 12: PID 2 running
Time 13: PID 2 finished
Time 13: PID 3 running
Time 14: PID 3 running
Time 15: PID 3 finished
Time 15: PID 1 running
Time 16: PID 1 running
Time 17: PID 1 running
Time 18: PID 1 running
Time 19: PID 1 finished

PID  Arrival  Start  End  Running  Waiting
0         0       0    12     12       0
1         2      15    19      4      13
2         3      12    13      1       9
3         4      13    15      2       9

Average Waiting Time: 7.75

```

RR:

```

matthew@LAPTOP-9I9622SL:~/CS4420/instruction-cs4420/Homework 4$ ./prog input.1 RR 5
RR:
Time 0: PID 0 running
Time 1: PID 0 running
Time 2: PID 0 running
Time 3: PID 0 running
Time 4: PID 0 running
Time 5: PID 1 running
Time 6: PID 1 running
Time 7: PID 1 running
Time 8: PID 1 running
Time 9: PID 1 finished
Time 9: PID 2 running
Time 10: PID 2 finished
Time 10: PID 3 running
Time 11: PID 3 running
Time 12: PID 3 finished
Time 12: PID 0 running
Time 13: PID 0 running
Time 14: PID 0 running
Time 15: PID 0 running
Time 16: PID 0 running
Time 17: PID 0 running
Time 18: PID 0 running
Time 19: PID 0 finished

PID  Arrival  Start  End  Running  Waiting
0         0       0    19     12       7
1         2       5     9      4       3
2         3       9    10      1       6
3         4      10    12      2       6

Average Waiting Time: 5.5

```

FCFS:

```

matthew@LAPTOP-9I9622SL:~/CS4420/instruction-cs4420/Homework 4$ gcc -Wall -o prog prog.c
matthew@LAPTOP-9I9622SL:~/CS4420/instruction-cs4420/Homework 4$ ./prog input.1 FCFS
FCFS:
Time 0: PID 0 running
Time 1: PID 0 running
Time 2: PID 0 running
Time 3: PID 0 running
Time 4: PID 0 running
Time 5: PID 0 running
Time 6: PID 0 running
Time 7: PID 0 running
Time 8: PID 0 running
Time 9: PID 0 running
Time 10: PID 0 running
Time 11: PID 0 running
Time 12: PID 0 finished
Time 12: PID 1 running
Time 13: PID 1 running
Time 14: PID 1 running
Time 15: PID 1 running
Time 16: PID 1 finished
Time 16: PID 2 running
Time 17: PID 2 finished
Time 17: PID 3 running
Time 18: PID 3 running
Time 19: PID 3 finished

PID  Arrival  Start  End  Running  Waiting
0      0         0     12    12       0
1      2        12    16     4      10
2      3        16    17     1      13
3      4        17    19     2      13

Average Waiting Time: 9

```